
Review Sentimental Classifier

Project Id: AIML-16

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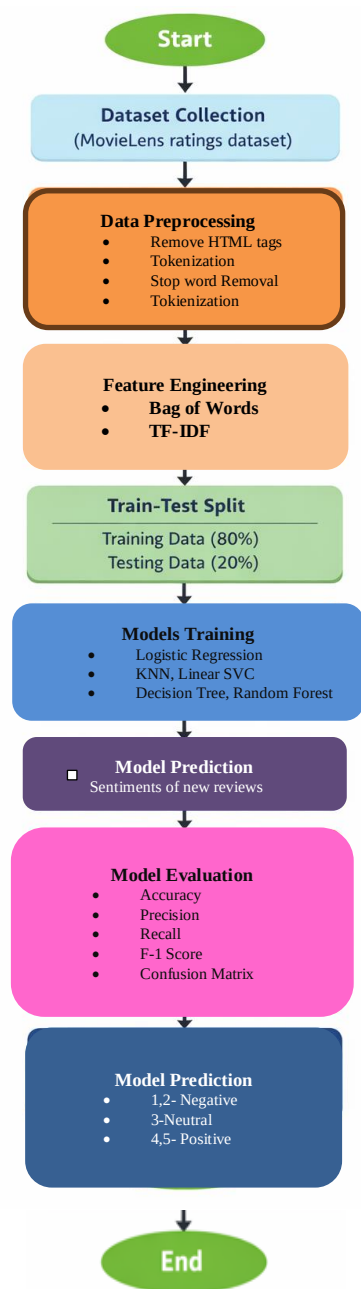
1. Problem Description

The project focuses on building an **end-to-end machine learning system** that predicts the **sentiment of product reviews**. The system classifies reviews into three categories: **Positive, Negative, and Neutral**.

The system takes raw textual data as input and processes it using **Natural Language Processing (NLP)** techniques. After preprocessing and feature extraction, machine learning models are trained to understand patterns in the data and predict sentiments accurately.

This system has real-world applications in **e-commerce platforms**, where it helps businesses analyze customer feedback, improve products, and enhance user satisfaction.

2. Architecture



3. Models Used(Applied/ Planned) Give a brief introduction of the models used

- **Logistic Regression**

A statistical model used for classification tasks. It works well for text classification and provides fast and interpretable results.

- **K-Nearest Neighbours (KNN)**

A distance-based algorithm that classifies a review based on similarity with neighboring data points.

- **Linear Support Vector Machine (SVM)**

A powerful classifier that finds an optimal boundary (hyperplane) between different sentiment classes. It performs well in high-dimensional text data.

- **Decision Tree**

A tree-based classification algorithm that splits the dataset into branches based on feature values. Each node represents a decision rule, each branch represents an outcome. Leaf nodes represent final class labels.

- **Random Forest**

A Random Forest is a machine learning algorithm that builds many Decision Trees. Each tree gives its own prediction. Final answer is made by majority voting(classification).

4. Evaluation metrics used

- **Accuracy**
Measures the overall correctness of predictions.
- **Precision**
Indicates how many predicted positive results are actually correct.
- **Recall**
Measures how well the model identifies all relevant cases.
- **F1-Score**
Balances precision and recall, useful for uneven datasets.
- **Confusion Matrix**
Shows correct vs incorrect predictions for each class.

5. Preliminary Results and Visualizations

```
Fitting 3 folds for each of 40 candidates, totalling 120 fits
=====
LOGISTIC REGRESSION – Best Hyperparameters
=====
{'C': 10, 'max_iter': 200, 'penalty': 'l1', 'solver': 'liblinear'}
Best CV Accuracy : 0.7168
Test Accuracy    : 0.7121

Classification Report:
      precision    recall  f1-score   support

   Negative      0.73      0.71      0.72      1705
    Neutral      0.62      0.62      0.62      1706
    Positive      0.78      0.80      0.79      1706

 accuracy          0.71          0.71          0.71      5117
  macro avg       0.71      0.71      0.71      5117
 weighted avg     0.71      0.71      0.71      5117
```

Fig.1

Fig. 2

```
=====
K-NEAREST NEIGHBORS – Best Hyperparameters
=====
{'metric': 'euclidean', 'n_neighbors': 15, 'p': 1, 'weights': 'uniform'}
Best CV Accuracy : 0.3408
Test Accuracy    : 0.3393

Classification Report:
      precision    recall  f1-score   support

   Negative      0.34      0.41      0.37      1705
    Neutral      0.33      0.31      0.32      1706
    Positive      0.34      0.29      0.31      1706

 accuracy          0.34          0.34          0.34      5117
  macro avg       0.34      0.34      0.34      5117
 weighted avg     0.34      0.34      0.34      5117
```

```

Fitting 3 folds for each of 3 candidates, totalling 9 fits

=====
SVM (LinearSVC) – Best Hyperparameters
=====
Best Params   : {'C': 0.1}
CV Accuracy   : 0.3334
Test Accuracy: 0.3334

      precision    recall  f1-score   support

   Negative      0.00      0.00      0.00      1705
    Neutral      0.33      1.00      0.50      1706
    Positive      0.00      0.00      0.00      1706

 accuracy          0.33      5117
 macro avg         0.11      0.33      0.17      5117
 weighted avg      0.11      0.33      0.17      5117

```

Fig. 3

```

Fitting 3 folds for each of 72 candidates, totalling 216 fits

=====
DECISION TREE – Best Hyperparameters
=====
Best Params   : {'criterion': 'gini', 'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 2}
CV Accuracy   : 0.5717
Test Accuracy: 0.5845

      precision    recall  f1-score   support

   Negative      0.59      0.58      0.58      1705
    Neutral      0.53      0.55      0.54      1706
    Positive      0.63      0.63      0.63      1706

 accuracy          0.58      5117
 macro avg         0.59      0.58      0.58      5117
 weighted avg      0.59      0.58      0.58      5117

```

Fig. 4

Fig. 5

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=====
RANDOM FOREST - Best Hyperparameters
=====
Best Params : {'criterion': 'gini', 'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 5, 'n_estimators': 200}
CV Accuracy : 0.7136
Test Accuracy: 0.7149

      precision    recall  f1-score   support

   Negative       0.73     0.70     0.72     1705
    Neutral       0.65     0.67     0.66     1706
    Positive       0.77     0.77     0.77     1706

 accuracy          0.71     0.71     5117
 macro avg         0.72     0.71     0.72     5117
weighted avg         0.72     0.71     0.72     5117
```

6. References:

Books:

- "Python Machine Learning" by Sebastian Raschka: A comprehensive guide that covers the implementation of Scikit-learn model and sentiment analysis workflows.
- "Natural Language Processing with Python" by Steven Bird: The definitive book for learning the NLTK library and text preprocessing techniques.
- Scikit-learn Official Documentation: scikit-learn.org—The best resource for understanding the parameters of Logistic Regression, SVM, and evaluation metrics.
- Kaggle Datasets and Kernels: kaggle.com— An excellent source for finding "Product Review" datasets and seeing how other data scientists handle text cleaning.